

Hearst OS — Architecture Technique

Plateforme IA Multi-Agents Enterprise | v2.0 | 2026

Mission

Hearst OS est le **système nerveux numérique** de l'entreprise moderne — un cerveau IA multi-agents qui comprend, apprend, et agit en autonomie supervisée.

Architecture Globale

EDGE	CDN Vercel → WAF / Rate Limiting / DDoS
GATEWAY	REST + SSE / WebSocket / JWT RS256 + RBAC
API	Next.js 15 API Routes / Edge Functions
AI CORE	Orchestrateur → Runtime → Prompt → Memory → RAG
DATA	PostgreSQL / Pinecone / Redis / S3

18 Microservices

SERVICE	RÔLE	TECH
API Gateway	Routing, sécurité	Next.js Edge
Auth Service	JWT, OAuth2, MFA	Supabase Auth
User Service	Profils, orgs	PG + RLS
Billing Service	Subscriptions	Stripe + BullMQ
AI Orchestrator	Intent, routing	Vercel AI SDK
Agent Runtime	Execution sandbox	Node.js VM
Prompt Engine	Templates, A/B	Langfuse
Memory Service	Short/Long/Vector	PG + Pinecone
Embedding Service	Vectorisation	OpenAI + Queue
Retrieval Service	Hybrid search	Pinecone
Workflow Engine	DAG execution	BullMQ + PG
Notification	Multi-canal	Resend + Slack
Analytics	Events, funnels	Segment + Grafana
Admin	Config, audit	Next.js
File Processing	OCR, chunking	Multer + Tesseract
Realtime	WebSocket	Socket.io
Vector DB	ANN search	Pinecone
Observability	Traces, metrics	Langfuse + Sentry

Stack Technique

COUCHE	TECHNOLOGIE	POURQUOI
Frontend	Next.js 15 + React 19 + Tailwind 4	SSR, RSC, design system
State	Zustand + Immer	Minimal, TS-first
Desktop	Electron + Vite	Codebase partagée
Auth	Supabase Auth	SOC2, RLS, MFA
AI SDK	Vercel AI SDK	Multi-LLM, streaming
Vector DB	Pinecone Serverless	<50ms, hybrid
Cache/Queue	Redis (Upstash)	Serverless, pub/sub
Storage	AWS S3 / R2	Durabilité 11 9s
Workers	BullMQ	Retry, DLQ
Observability	Langfuse + Prometheus + Grafana + Sentry	LLM tracing
CI/CD	GitHub Actions + Vercel	Edge deploy
IaC	Terraform + Docker	Reproductible

Cas d'Use & Valeur

CAS	AGENTS	GAIN
Analyse financière	Analyste + Data + Notif	-80%
Génération contenu	Rédacteur + Relecteur	×5
Recherche intelligente	Retrieval + Synthèse	95%+
Ordonnancement	Planning + Calendar	ZERO OUBLI
Support technique	Support + Tech	-60%
Veille stratégique	Veille + Analyste	TEMPS RÉEL

Coûts & Scaling

	\$2.6K 10K MAU	\$12K 100K MAU	\$80K 1M MAU	\$500K 10M MAU
POSTE	MENSUEL			
Infra (Vercel + Supabase + Pinecone + Redis + S3)	~\$1,436			
LLM (OpenAI + Anthropic)	~\$1,200			
TOTAL	~\$2,636			

Hearst OS — Détail Technique

Agents | LLM | RAG | Sécurité | Infrastructure

🤖 Multi-Agent System

🔧 Orchestrateur — Commandeur

Patterns: Router | Chainier | Parallel | Evaluator-Optimizer

👤 Analyste

🔍 Rechercheur

💻 Codeur

✍️ Rédacteur

📅 Planificateur

✅ Vérificateur

⚡ Execution Loop



🛡️ Sécurité — Defense in Depth

🌐 EDGE	WAF + DDoS + Bot (Cloudflare + Arcjet)
🔒 TRANSPORT	TLS 1.3 + mTLS interne
🔑 AUTH	JWT RS256 + OAuth2 + MFA
🔐 AUTHZ	RBAC + ABAC + PostgreSQL RLS
✅ INPUT	Zod schemas + Regex sanitization
🛡️ AI SAFE	Prompt injection + Jailbreak + PII
🔒 DATA	AES-256 at rest + Field encryption
📄 AUDIT	Immutable signed logs + GDPR

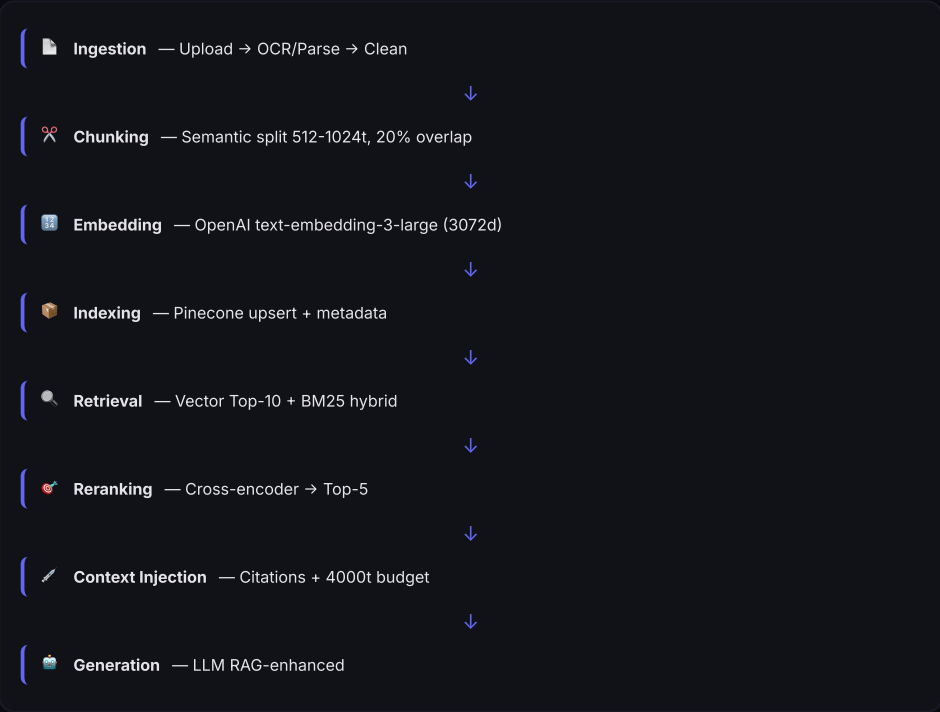
📊 Monitoring

DASHBOARD	MÉTRIQUES	AUDIENCE
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🧠 LLM Routing Intelligent

TIER	MODÈLE	INPUT	USE CASE
PREMIUM	GPT-4o / Claude 3.5	\$2.5-3/M	Raisonnement complexe
PREMIUM	Claude 3.5 Sonnet	\$3/M	Long contexte (200K)
STANDARD	GPT-4o-mini	\$0.15/M	Tâches générales (80%)
STANDARD	Claude 3 Haiku	\$0.25/M	Réponses rapides
MULTIMODAL	Gemini 1.5 Pro	\$3.5/M	1M contexte, multimodal
FALLBACK	Gemini Flash / Local	\$0.35/M	Backup, offline

📁 Pipeline RAG Complet



☁️ Infrastructure Cloud

COUCHE	TECHNOLOGIE	SCALING
Edge	Vercel Edge Network	Global CDN
Compute	Next.js Serverless + ECS Fargate	Auto-scale